
Enriched Physical Representations by predicting missing physical attribute latents

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Abstract

Understanding the spatiotemporal dynamics of active matter systems remains a fundamental challenge at the intersection of biophysics and machine learning. Prior work has explored applying Video Joint Embedding Predictive Architectures (VJEPA) to datasets such as Well, motivated by the strong correlation between video representations and underlying spatiotemporal dynamics. While VJEPA demonstrates promising performance, it falls short compared to more specialized approaches. A key limitation of JEPA-style methods is their lack of explicit incentives to capture complex physical structure, such as the governing dynamics of the system. In this work, we address this gap by augmenting the JEPA objective to explicitly predict latent representations corresponding to missing physical information. Specifically, we investigate masking different physical channels and fields, and analyze how reconstructing these masked components influences the learned representations. Our findings suggest that masking physical fields and requiring the predictor to infer them from complementary signals encourages the model to learn more physically meaningful features. This results in improved individual contribution of each field to the learned representation. We evaluate our approach using linear probing and k-nearest neighbors (kNN), and provide detailed ablation studies over multiple masking strategies on 20 % of the dataset.

1. Introduction

Active matter systems — ranging from bacterial suspensions and microtubule assemblies to collections of self-propelled synthetic particles — exhibit rich, out-of-equilibrium spatiotemporal dynamics driven by the interplay of activity, alignment, and hydrodynamic interactions (1; 2). Modeling

such systems typically requires tracking high-dimensional particle distributions via kinetic theories, or solving coarse-grained partial differential equations over physical fields such as concentration c and orientation tensor $\mathbf{D}$. Despite these simplifications, accurately capturing the evolution of these systems remains computationally expensive and often relies on approximate closure models.

A fundamental question is whether the complex dynamics of active matter systems can be encoded into compact representations that preserve the underlying physical structure. Recent advances in self-supervised learning, particularly Joint-Embedding Predictive Architectures (JEPA) (5; 6), offer a promising direction. By learning to predict representations of future observations in latent space, JEPA-based models have demonstrated strong performance in video understanding tasks, where temporal correlations provide rich supervisory signals.

Motivated by the analogy between video sequences and evolving physical fields, prior work has explored applying Video-JEPA (VJEPA) models to large-scale simulation datasets such as *The Well* (4). While these approaches leverage spatiotemporal structure effectively, they do not explicitly enforce learning of the underlying physical dynamics. In particular, standard JEPA objectives focus on predicting latent representations without encouraging the model to reason about missing or unobserved physical quantities, limiting their ability to capture deeper physical dependencies across fields.

In this work, we address this limitation by introducing a physically motivated masking strategy within the JEPA framework. Specifically, we explicitly mask selected physical channels and fields, and train the predictor to reconstruct their latent representations from the remaining observed information. This forces the model to infer missing physical quantities and encourages stronger cross-field interactions in the learned representations.

We systematically explore different masking strategies across physical fields and analyze their impact on representation quality. Our results show that masking physical information leads to more disentangled and physically meaningful representations, improving the individual contribution of each field. We evaluate our approach using lin-

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ear probing and k-nearest neighbors (kNN) on downstream physical parameter prediction tasks, and provide detailed ablation studies over multiple design choices on a subset of the dataset.

Overall, our findings highlight that explicitly incorporating physical structure into self-supervised objectives is crucial for learning representations that go beyond spatiotemporal correlation and capture the underlying dynamics of active matter systems.

2. Dataset: Active Matter

2.1. Physical Background

The `active_matter` dataset models the dynamics of N rod-like active particles of aspect ratio $b \ll 1$ immersed in a fluid within a periodic square domain of length $L = 10$. The full physical system is described by a continuum distribution function $\psi(\mathbf{x}, \mathbf{p}, t)$, tracking the center-of-mass position $\mathbf{x}$ and orientation vector $\mathbf{p}$ of each particle at time t . Its evolution is governed by the *Smoluchowski equation*, which conserves particle number and reads

$$\partial_t \psi + \nabla_{\mathbf{x}} \cdot \mathbf{J}_{\mathbf{x}} + \nabla_{\mathbf{p}} \cdot \mathbf{J}_{\mathbf{p}} = 0, \quad (1)$$

where the conformational fluxes $\mathbf{J}_{\mathbf{x}}$ and $\mathbf{J}_{\mathbf{p}}$ encode particle advection, translational diffusion (d_T), rotational diffusion (d_R), and steric alignment interactions with strength ζ . This is coupled to the Stokes flow equations:

$$\nabla \cdot \boldsymbol{\sigma} = \nabla P, \quad \nabla \cdot \mathbf{u} = 0, \quad (2)$$

where the stress tensor $\boldsymbol{\sigma}$ contains contributions from the active dipole (strength α), particle rigidity, and steric torques. The zeroth and second moments of ψ with respect to $\mathbf{p}$ yield the concentration field c and the nematic orientation tensor $\mathbf{D} = \langle \mathbf{p}\mathbf{p} \rangle_c$, respectively.

The coarse-grained description, obtained by tracking only c and $\mathbf{D}$, requires a *closure* for the fourth-order moment tensor $\mathbf{S} = \langle \mathbf{p}\mathbf{p}\mathbf{p}\mathbf{p} \rangle_c$ that appears in the moment equations. The two key physical parameters governing the dynamics are:

[noitemsep]

- α : the dimensionless **active dipole strength**, controlling the magnitude of activity-driven fluid stresses (5 discrete values).
- ζ : the dimensionless **steric alignment strength**, driving the isotropic-to-nematic phase transition in dense suspensions (9 discrete values).

There are $5 \times 9 = 45$ unique (α, ζ) combinations in total. Raw kinetic simulations use 256^2 spatial modes and 256

orientational modes with a pseudo-spectral discretization and SBDF2 time-stepping ($\Delta t = 4 \times 10^{-4}$), running for 81 time steps.

2.2. Dataset Structure

The processed `active_matter` dataset is hosted on HuggingFace (`polymathic-ai/active_matter`) and constitutes approximately **52 GB** of data, making it the most accessible entry in *The Well*. Its structure is summarized in Table 1.

Table 1. Summary of the `active_matter` dataset.

Property	Value
Total size	≈ 52 GB
Temporal resolution	16 frames per sample (from 81 raw time steps)
Spatial resolution	224×224 (cropped from 256×256)
Physical channels	11
Training samples	8,750
Validation samples	1,200
Test samples	1,300
Physical parameters	α (5 values), ζ (9 values) — 45 combinations

2.3. Physical Fields (Channels)

Each sample is a tensor of shape $[16, 224, 224, 11]$ (time $\times$ height $\times$ width $\times$ channels). The 11 channels decompose into four physically meaningful fields:

[noitemsep]

- **Concentration scalar field** (c , 1 channel): the zeroth moment of ψ , representing local particle density. Under the uniform-concentration assumption used in simulations, $c(\mathbf{x}, t) = 1$; deviations encode non-trivial spatial structure.
- **Velocity vector field** ($\mathbf{u}$, 2 channels: u_x, u_y): the activity-driven fluid velocity satisfying the Stokes equations. Captures large-scale coherent flows and active turbulence.
- **Orientation tensor field** ($\mathbf{D}$, 4 channels: $D_{11}, D_{12}, D_{21}, D_{22}$): the symmetric second-order nematic order parameter, encoding the mean alignment direction and degree of order of the particles.
- **Strain-rate tensor field** ($\mathbf{E}$, 4 channels: $E_{11}, E_{12}, E_{21}, E_{22}$): the symmetric part of the velocity gradient, $\mathbf{E} = (\nabla \mathbf{u} + (\nabla \mathbf{u})^\top)/2$, characterizing local deformation rates in the fluid.

Together, these 11 channels provide a complete coarse-grained description of the active suspension at each time step. The temporal sequence of 16 frames captures the full range of dynamical phenomena — including defect

formation, active turbulence, and the isotropic-to-nematic transition — that are central to understanding the system’s physics.

2.4. Dataset Analogy to Video

A key insight that motivates the choice of architecture is the structural analogy between this dataset and video data. Just as a video clip is a sequence of T RGB images (3 channels), each `active_matter` sample is a sequence of 16 spatial grids with 11 physical channels evolving over time. This analogy is not merely superficial: the spatiotemporal correlations present in physical simulations — coherent structures, advection, wave propagation — are closely related to the motion and temporal coherence that video understanding models are designed to exploit. This motivates the direct adaptation of architectures such as Video JEPA and Video-MAE to the physical simulation domain.

3. Method

3.1. JEPA

Joint-Embedding Predictive Architectures (JEPA) (8) learn representations by predicting one view from another in a shared embedding space, avoiding reconstruction and contrastive objectives. I-JEPA (5) applies this to images using a student–teacher setup with an EMA target encoder. V-JEPA (6) extends this framework to video by predicting masked spatiotemporal regions, encouraging modeling of temporal dependencies via aggressive 3D masking. Recent extensions (7) incorporate rollout prediction for multi-step forecasting in latent space, connecting JEPA to latent world models that capture underlying system dynamics.

Formally, given an input x and a masking operator $\mathcal{M}$ that produces a context $x_c = \mathcal{M}(x)$ and target regions x_t , JEPA learns encoders f_θ (context) and $f_{\bar{\theta}}$ (target, updated via EMA) that map inputs to latent representations. A predictor g_ϕ maps the context embedding to the target embedding:

$$\mathbf{z}_c = f_\theta(x_c), \quad \mathbf{z}_t = f_{\bar{\theta}}(x_t), \quad \hat{\mathbf{z}}_t = g_\phi(\mathbf{z}_c, \mathbf{p}_t), \quad (3)$$

where $\mathbf{p}_t$ encodes positional information of the masked region. The training objective minimizes a feature-space regression loss

$$\mathcal{L} = \mathbb{E}_{x \sim \mathcal{D}} \|\hat{\mathbf{z}}_t - \text{sg}(\mathbf{z}_t)\|_2^2, \quad (4)$$

with $\text{sg}(\cdot)$ denoting stop-gradient. For temporal data, the predictor can be iteratively applied to produce multi-step rollouts $\hat{\mathbf{z}}_{t+1} = g_\phi(\hat{\mathbf{z}}_t, \mathbf{p}_{t+1})$, yielding a learned latent dynamical system trained via the same objective over future steps.

For the `active_matter` dataset, the V-JEPA framework is adapted as follows. Each simulation sample (shape

[16, 224, 224, 11]) is tokenized into 3D spacetime patches, with the 11 physical channels playing the role of the RGB (and beyond) channel dimension. The ViT-based encoder processes these tokens to produce a latent vector $\mathbf{z} \in \mathbb{R}^d$ per token, which is then pooled into a global representation. The EMA target encoder ensures stable training targets and guards against representation collapse.

A critical constraint of this project is that **no physical parameter labels** (α or ζ) are used during representation learning. Instead, the model is evaluated post-hoc by freezing the encoder and fitting either a single linear layer or a kNN regressor on the pooled representations to predict the continuous (z-score normalized) values of α and ζ , measured via MSE. A well-trained JEPA model should produce representations that are predictive of these parameters, since they govern the large-scale spatiotemporal structure — alignment, defect density, turbulent intensity — captured in the learned latent space.

The baseline JEPA model for this task contains approximately 3.3M parameters (~ 2.5 M for the encoder and ~ 0.8 M for the predictor), yet can consume nearly 100 GB of VRAM at a batch size of 8 due to the high-dimensional spatiotemporal activations — a stark reminder of the computational challenges unique to this domain compared to standard video benchmarks.

3.2. Why JEPA Failed

JEPA predicts the future in the latent space, which eliminates not predictable elements from influencing the prediction. A major flaw in current JEPA based approaches is that there is no regularization that prevents not learning hard to decipher but predictable features, like the physics of the system.

3.3. Field Masking

Let $\mathbf{x}_t \in \mathbb{R}^{C \times T \times H \times W}$ denote the context input at timestep t , where $C = 11$ channels correspond to four physical fields: concentration c (1 channel), velocity $\mathbf{U}$ (2 channels), orientation tensor $\mathbf{D}$ (4 channels), and strain-rate tensor $\mathbf{E}$ (4 channels). At each training step, we sample a field index $k \sim \text{Uniform}\{0, 1, 2, 3\}$ and zero its corresponding channels:

$$\tilde{\mathbf{x}}_t^{(k)} = \mathbf{x}_t \odot (\mathbf{1} - \mathbf{m}_k) \quad (5)$$

where $\mathbf{m}_k \in \{0, 1\}^C$ is a binary mask with ones at the channel indices of field k and zeros elsewhere. The masked input $\tilde{\mathbf{x}}_t^{(k)}$ is passed to the encoder f_θ , while the prediction target remains the full next-frame encoding:

$$\mathcal{L}_{\text{mask}} = \mathcal{L}_{\text{Cost}}\left(g_\phi\left(f_\theta\left(\tilde{\mathbf{x}}_t^{(k)}\right)\right), f_\theta(\mathbf{x}_{t+1})\right) \quad (6)$$

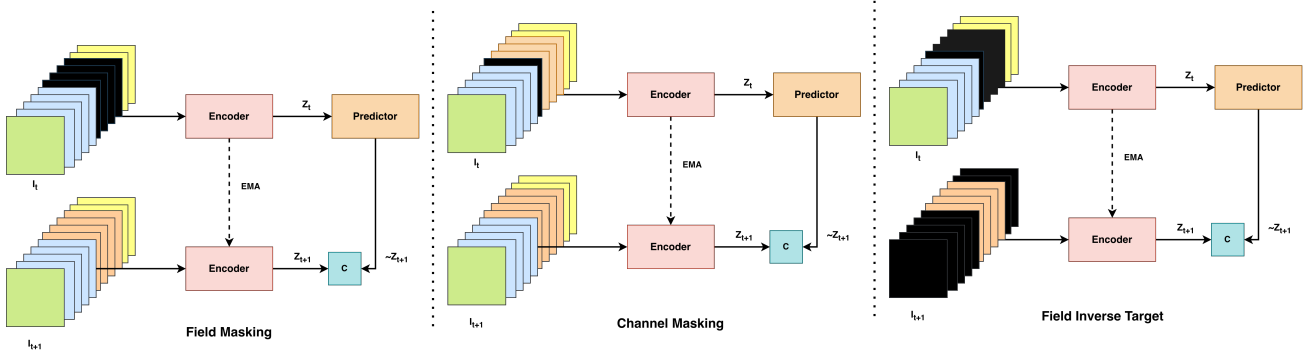


Figure 1. All Masking strategies explored in this report. Field Masking masked a physical attribute like velocity at a designated time step. Channel Masking masked only a random channel. Field Inverse Target in addition to Field Masks also masked the target inversely to force the predictor to predict the future latent of the masked field.

where g_ϕ denotes the predictor network. Since k is resampled independently at every training step, the encoder cannot specialize to any fixed subset of fields — it must maintain representations sufficient to predict the complete next-frame state from any partial field observation. The target is kept as the full next-frame representation rather than the masked field alone; predicting across field modalities and time simultaneously was found empirically to be intractable given the predictor capacity used (Section 6). Field masking is applied at 50% of training steps, alternating with standard unmasked JEPA steps to preserve stable temporal prediction as the primary objective.

3.4. Smooth ℓ_1 vs. MSE in feature space.

Although the original I-JEPA objective in Eq. (??) is written as an ℓ_2 distance, V-JEPA in practice replaces this with a smooth- ℓ_1 (Huber) loss between the predicted and EMA-teacher embeddings. The reason is that the EMA teacher is an inherently noisy regression target: its weights lag the student by a momentum schedule, and on a small fraction of patches it produces high-variance outlier embeddings — especially early in training before the student has converged. Under MSE, those outliers contribute quadratic gradients that disproportionately drag the student toward the teacher’s noise on a few patches per batch. Smooth ℓ_1 is locally ℓ_2 for $|x| < \beta$ (so it retains MSE’s clean convergence on the bulk of small errors) but globally ℓ_1 for large $|x|$ (so the gradient on outlier targets is bounded). Together with the fact that EMA-teacher V-JEPA has no variance/ covariance regulariser to absorb instabilities, this bounded-gradient behaviour acts as an implicit safeguard against drift toward degenerate solutions and is the choice we adopt in our implementation.

4. Experiments

4.1. Downstream Parameter Estimation

We report the validation MSE of frozen-encoder α and ζ regression on the full `active_matter` training split, evaluated under both a single linear probe and a $k=20$ kNN regressor on z -scored targets (means $[-3.0, 9.0]$, stds $[1.41, 5.16]$ for α, ζ respectively). The supervised baseline is included as an end-to-end upper bound; all other rows freeze the encoder learnt during self-supervised pretraining. Linear results are reported at the final training epoch (100); kNN results use $k=20$ from the kNN sweep.

Table 2. Linear-probe validation MSE on z -scored (α, ζ) (full train set, 100 epochs). Per-parameter losses were not logged for the end-to-end supervised run. **Bold**: best frozen-encoder probe.

Encoder	α	ζ	avg
Baseline JEPA (VICReg)	0.613	0.724	0.669
EMA-temporal V-JEPA	0.034	0.781	0.407
Field-mask EMA V-JEPA	0.089	0.876	0.482
Supervised (end-to-end)	—	—	0.055

Table 3. kNN ($k=20$) validation MSE on z -scored (α, ζ) (full train set). **Bold**: best frozen self-supervised encoder.

Encoder	α	ζ	avg
Baseline JEPA (VICReg)	0.996	0.988	0.992
EMA-temporal V-JEPA	0.683	0.859	0.771
Field-mask EMA V-JEPA	0.077	0.759	0.418
Supervised encoder $\rightarrow$ kNN	0.002	0.111	0.056

Discussion. Two trends are visible across both probes. First, replacing the original shared-weights VICReg target encoder with an EMA teacher (EMA-temporal V-JEPA)

cleanly improves over the baseline: linear-probe avg MSE drops from $0.669 \rightarrow 0.407$ and α recovery in particular jumps from $0.613 \rightarrow 0.034$, consistent with the EMA predictive objective allocating more representational capacity to the dynamical structure that α governs. Second, layering field masking on top of the EMA recipe further *improves* the kNN result substantially ($0.771 \rightarrow 0.418$ avg, $0.683 \rightarrow 0.077$ on α), though it is slightly worse than plain EMA under the linear probe ($0.407 \rightarrow 0.482$). The interpretation is that field masking re-organises the representation toward more locally clustered, label-aware geometry — which a non-linear neighbourhood-based decoder can exploit better than a single linear projection. The end-to-end supervised baseline (which trains the encoder *using* the labels) sits well below all frozen-encoder probes, as expected.

4.2. k-Nearest Neighbor Regression Probe

To evaluate the quality of learned representations without any gradient updates to the backbone, we employ a k -nearest neighbors (kNN) regression probe on frozen encoder features. After pre-training, the context encoder is fixed and used to extract embeddings for all training and validation samples. Specifically, each input clip of 16 context frames is passed through the frozen encoder, and the resulting spatial feature map is reduced to a single vector via global average pooling.

Both the feature vectors and target labels (α , ζ) are standardized (zero-mean, unit-variance) using statistics computed from the training set. A `KNeighborsRegressor` with $k=10$ and cosine distance is then fit on the training embeddings and used to predict the physical parameters on the held-out validation set. Performance is reported as mean squared error (MSE) on the z -scored targets for each parameter independently. No labels are used during pre-training; the kNN probe is applied only at evaluation time to measure how linearly and locally separable the representations are with respect to the underlying physics parameters.

5. Analysis

5.1. t-SNE Visualization of kNN Representations

To qualitatively assess the structure of learned representations, we project the frozen context encoder embeddings onto two dimensions using t-distributed stochastic neighbor embedding (t-SNE) and color each point by the underlying physical parameters ζ (alignment strength) and α (dipole strength). The probe uses $k=10$ nearest neighbors, and no gradient updates are made to the backbone during evaluation.

The t-SNE layout in Figure 2 reveals a *spiral-arc* structure, where each elongated streak corresponds to a single simulation trajectory. The dense central cluster represents

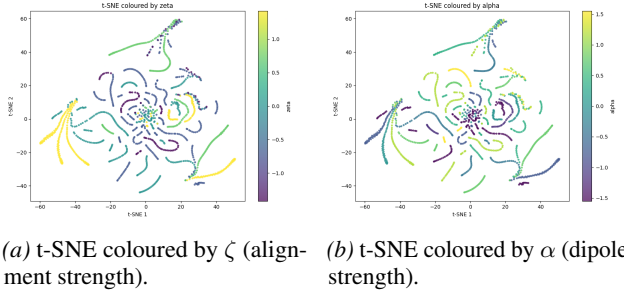


Figure 2. t-SNE projections of frozen JEPA encoder embeddings on the `active_matter` dataset ($k=10$). Each arc corresponds to a single simulation trajectory evolving over 16 time steps. Color encodes the value of the respective physical parameter.

early-time or near-isotropic states, where dynamics across different parameter settings are largely indistinguishable. Radial distance from this center reflects the degree of departure from the isotropic regime toward active turbulence or nematic order.

As shown in Figure 2a, trajectories are colored uniformly within individual arcs when mapped by ζ , indicating that the encoder has learned to cluster simulations by their steady-state nematic order. Trajectories with high ζ (strong alignment) occupy spatially distinct peripheral regions, consistent with the physically distinct flow patterns of the nematic phase. In contrast, Figure 2b exhibits smooth color *gradients* along individual arcs, suggesting that α (dipole activity) modulates the temporal evolution of system dynamics rather than its steady-state cluster membership. Together, these plots confirm that the JEPA representations are physically structured and disentangled with respect to both governing parameters of the `active_matter` system, without any explicit supervision.

5.2. Field Importance via Zero-Masking

To assess which physical fields contribute most to the learned representations, we employ a zero-masking ablation study on the pretrained encoder.

Setup. The active-matter dataset comprises 11 channels organized into four physical field groups: *Concentration* (1 channel), *Velocity* (2 channels), *Orientation* (4 channels), and *Strain* (4 channels).

Method. For each field f , all of its associated channels are zeroed out across every input sample. The original input x and the masked variant $x^{(f)}$ are each passed independently through the frozen encoder g_θ , yielding latent representations $z_{\text{full}} = g_\theta(x)$ and $z_{\text{masked}}^{(f)} = g_\theta(x^{(f)})$. The importance of field f is then measured as the expected mean squared

error (MSE) between the two representations:

$$\mathcal{I}(f) = \mathbb{E}_x \left[\|z_{\text{full}} - z_{\text{masked}}^{(f)}\|^2 \right]. \quad (7)$$

A larger $\mathcal{I}(f)$ indicates that the encoder’s representation changes substantially upon removal of field f , implying greater model reliance on that field. To account for the varying channel dimensionality across fields, we report both the raw per-field MSE and a *channel-normalized* MSE, computed by dividing $\mathcal{I}(f)$ by the number of channels in field f .

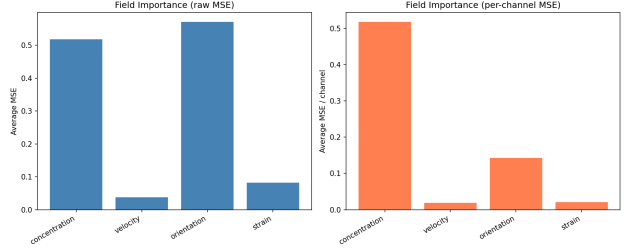
Implementation. The encoder is loaded from a pretrained checkpoint and kept frozen throughout evaluation (no gradient computation). The full training set is iterated once per evaluation run; for each mini-batch, masked variants for all four fields are constructed and processed in a sequential forward-pass loop. Results are aggregated, presented as a ranked table, and visualized as a bar chart.

5.3. Field Importance Ablation Results

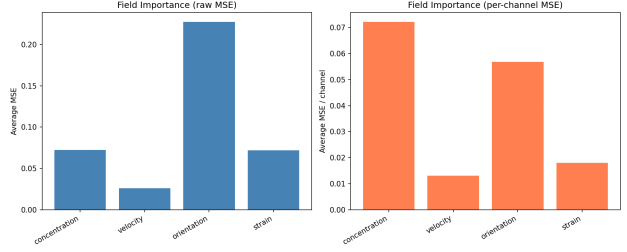
Both figures show field importance scores derived from zero-masking ablation — each physical field (*concentration*, *velocity*, *orientation*, *strain*) is individually zeroed out and the resulting MSE between the full and masked latent representations quantifies how much each field contributes to the encoder’s representation.

The **baseline model** shows that **orientation dominates** with the highest raw MSE (≈ 0.57), followed by concentration (≈ 0.52), while velocity and strain are comparatively minor (≈ 0.04 and ≈ 0.08); however, once normalized per channel, concentration leaps to the top ($\approx 0.52/\text{channel}$), surpassing orientation ($\approx 0.15/\text{channel}$), which reveals that orientation’s raw dominance is largely explained by its higher channel dimensionality rather than true per-channel informativeness.

The **field-masked model** tells a noticeably different story: the overall MSE magnitudes drop substantially across all fields (e.g., orientation’s raw MSE falls from ≈ 0.57 to ≈ 0.23 , and concentration from ≈ 0.52 to ≈ 0.07), suggesting the field-masking training strategy forces the encoder to distribute representations more evenly and become less sensitive to any single field being zeroed out. In the per-channel normalized view, concentration still leads ($\approx 0.072/\text{channel}$) but orientation now closely competes ($\approx 0.057/\text{channel}$) with a more balanced spread across all four fields — indicating that field masking successfully reduces over-reliance on individual fields and encourages a more robust, distributed latent representation.



(a) Field importance scores under zero-masking ablation on baseline model.



(b) Field importance scores under zero-masking ablation on the field masked model.

Figure 3. **Left:** Raw MSE between full and masked latent representations. **Right:** Per-channel normalized MSE, controlling for field dimensionality.

6. Ablation Studies

We isolate the contribution of each design decision in our method through a series of controlled ablations on the active matter dataset (20% training subset, full fine-tuning dataset). All variants share the same encoder architecture, VICReg loss coefficients, and training schedule unless otherwise noted. Fine-tuning uses a frozen encoder with a linear head for 100 epochs, reporting mean-squared-error (MSE) on physical parameter estimation averaged over α (active dipole strength) and ζ (alignment strength). We additionally report the *field importance uniformity* (FI uniformity) metric, defined as $1 - \sigma/\mu$ of the per-channel MSE across the four physical fields when each is individually masked from the encoder input (higher = more uniform field sensitivity).

6.1. Masking Token Type: Zero vs. Learnable

When a physical field is masked in the context input, its channels must be replaced by some placeholder value. Two natural choices exist: a fixed zero and a learned scalar parameter $\theta \in \mathbb{R}$ broadcast across the masked channel.

A *zero mask* is the simplest option — it introduces no additional parameters and makes a clear distinction between present and absent information, since zero is outside the natural range of normalised field values. A *learnable mask token* (following ?) allows the encoder to distinguish “this channel was masked” from “this channel genuinely has zero

value,” which may help in systems where zero is a meaningful physical state (e.g., zero velocity in a quiescent region). However, a learnable scalar introduces a small optimisation interaction: the token value is updated by gradients flowing through the encoder, so it converges toward whatever value minimises the loss rather than a physically neutral value. For active matter, the fields are normalised and zero is not a physically special value for any field, so the distinction between zero and a learned token is expected to be minor. We test both to confirm this and to determine whether the additional parameter provides any benefit.

Table 4. Effect of masking token type on representation quality and downstream parameter estimation. All runs use field-level masking at 50% of steps, no field-ID conditioning on the predictor.

Mask token	repr loss ↓	FT MSE ↓
Zero mask	1.770	0.639
Learnable mask	1.826	0.762

6.2. Field-ID Conditioning on the Predictor

When field masking is applied, the predictor receives a context embedding z_{context} derived from a partial observation — it does not inherently know which field was masked. One approach is to provide an explicit field-identity signal: a learned embedding $\mathbf{e}_k \in \mathbb{R}^D$ for the masked field index k is added to z_{context} before the predictor, implemented via `FieldConditionedPredictor` (Section ??). The motivation is that the predictor can specialise its computation per field, potentially making the prediction task easier and improving training stability.

However, this creates a tension with our theoretical goal. If the predictor is told which field was masked, it can route around a poorly structured encoder representation — the encoder no longer needs to produce a unified, field-agnostic embedding. By *removing* the field-ID signal, we force the predictor to use a single mapping for all masking patterns, which in turn requires the encoder to produce representations that are consistent regardless of which field is absent. This is precisely the field-agnosticism property we desire.

Empirically, we find that removing field-ID conditioning reduces the gap between the masked method’s representation loss and the baseline, suggesting the encoder learns more consistent representations without the routing cue. We ablate both settings to quantify this effect.

6.3. Masking Strategy: Channel, Field, and Inverse

The granularity and target of the masking objective are the central design decisions of our method. We compare three strategies against the unmasked JEPa baseline.

Channel masking zeros a single channel (out of 11) chosen

Table 5. Effect of field-ID conditioning on the predictor. *With z* : predictor receives an additive field embedding $\mathbf{e}_k$ indicating which field was masked. *Without z* : predictor receives only z_{context} , forcing the encoder to produce field-agnostic representations. All runs use zero masking, 50% masking rate. “repr loss gap” is the difference between the method’s val repr loss and the baseline val repr loss at the end of training (lower = closer to baseline = better).

Method	Rep MSE ↓	FT MSE ↓
With field z	1.770	0.639
Without field z	1.953	0.8377

uniformly at random. This is the finest-grained masking — it does not respect physical field boundaries, potentially splitting a tensor field (e.g., masking one component of the 4-channel orientation tensor $\mathbf{D}$ while leaving the other three visible). The motivation for testing this variant is to determine whether *physical* field boundaries are necessary or whether any input perturbation produces similar benefits.

Field masking zeros all channels of one complete physical field chosen uniformly at random from the four fields (c , $\mathbf{U}$, $\mathbf{D}$, $\mathbf{E}$). This respects the physical structure of the data — each masked unit is a coherent physical quantity — and is our primary proposed strategy. The prediction target is the full next-frame representation across all fields.

Inverse masking applies field masking to the context but also restricts the target: the target encoder sees only the masked field’s channels at $t+1$, forcing the predictor to bridge across both field modality and time simultaneously. This was motivated by the desire to explicitly condition the encoder on representing each individual field’s future state. However, as we show empirically, this cross-modal temporal prediction task is intractable for the small predictor capacity used — the representation loss plateaus immediately and never decreases, because the encoder has not yet learned to align representations of different field subsets. We include this variant as an ablation to document this failure mode and to motivate the simpler full-target formulation.

All masking variants use a 50% masking rate (applied every other step), zero mask token, and no field-ID conditioning. We report representation loss at the end of pretraining, field importance uniformity (FI uniformity), and fine-tuning MSE broken down by α and ζ to assess whether the masking strategy differentially benefits parameters encoded in different fields.

7. Statement of Contributions

7.1. Contributions Shashaank Ravichandran(sr8016)

1. k-Nearest Neighbor Regression Probe

Table 6. Comparison of masking strategies. *Baseline*: standard JEPA, no masking. *Channel*: single random channel zeroed. *Field*: all channels of one physical field zeroed, full next-frame target. *Inverse*: field-masked context, masked-field-only target at $t+1$. α and ζ MSE reported separately as our method is expected to improve α estimation most, given the identified underrepresentation of velocity (which encodes α) in baseline JEPA.

Method	Rep MSE ↓	FT MSE ↓
Field z	1.770	0.639
Channel z	1.760	0.692
Field Inverse Target	1.982	0.977

2. TSNE for KNN
3. Field Importance via Zero-Masking
4. Ablations

7.2. Contributions Akash Guna Royar Thangavelu(ar9799)

1. Field Mask, Channel Mask, Inverse Target theorization and implementation
2. Ablations
3. Field Importance Theorization

7.3. Contributions Kishlay Kumar (kk5047)

1. Baseline paper Replication
2. Implementation of end-to-end supervised fine tuned baseline
3. EMA encoder implementation

Impact Statement

Authors are **required** to include a statement of the potential broader impact of their work, including its ethical aspects and future societal consequences. This statement should be in an unnumbered section at the end of the paper (co-located with Acknowledgements – the two may appear in either order, but both must be before References), and does not count toward the paper page limit. In many cases, where the ethical impacts and expected societal implications are those that are well established when advancing the field of Machine Learning, substantial discussion is not required, and a simple statement such as the following will suffice:

“This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.”

The above statement can be used verbatim in such cases, but we encourage authors to think about whether there is content which does warrant further discussion, as this statement will be apparent if the paper is later flagged for ethics review.

References

- [1] S. Ramaswamy. The mechanics and statistics of active matter. *Annu. Rev. Condens. Matter Phys.*, 1:323–345, 2010.
- [2] M. C. Marchetti, J.-F. Joanny, S. Ramaswamy, T. B. Liverpool, J. Prost, M. Rao, and R. A. Simha. Hydrodynamics of soft active matter. *Reviews of Modern Physics*, 85:1143, 2013.
- [3] S. Maddu, et al. Data-driven closure models for active matter systems. *arXiv preprint*, 2024.
- [4] Polymathic AI. The Well: A large-scale collection of physical simulations. https://huggingface.co/datasets/polymathic-ai/active_matter, 2024.
- [5] M. Assran, Q. Duval, I. Misra, P. Bojanowski, P. Vincent, M. Rabbat, Y. LeCun, and N. Ballas. Self-supervised learning from images with a joint-embedding predictive architecture. *CVPR*, 2023.
- [6] A. Bardes, Q. Garrido, J. Ponce, X. Chen, M. Rabbat, Y. LeCun, M. Assran, and N. Ballas. V-JEPA: Latent video prediction for visual representation learning. *arXiv:2404.08471*, 2024.
- [7] A. Bardes, et al. V-JEPA 2: Self-supervised video models enable understanding, prediction and planning. *arXiv*, 2025.
- [8] Y. LeCun. A path towards autonomous machine intelligence. *OpenReview preprint*, 2022.

A. You *can* have an appendix here.

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